the highest level are the formal ISO standards (for example, ISO 30173 (https://www.iso.org/standard/81442.html), ISO 30194 (https://www.iso.org/standard/53314.html)) that provide a high-level framework and conceptual model for digital twins themselves. Understanding this tiered approach is important because it allows you to identify and implement the most critical standards first, laying a solid foundation before tackling the more abstract, higher-level frameworks. A well-considered strategy for standards implementation ensures not only technical compatibility but also long-term scalability and future-proofing of your digital twin.

5.5.2 Practical standards assessment

Navigating the world of any standards requires a careful assessment of their practical value and associated costs. Adopting a standard isn't just a technical decision but a strategic investment that can bring significant benefits but also comes with hidden costs, particularly related to compliance and audits. This section provides a practical framework for evaluating whether a standard is worth adopting based on its ability to solve a real-world problem or its long-term relevance in the industry.

THE TRUE COST OF STANDARDS ADOPTION

The cost of adopting a standard goes beyond initial implementation. It includes:

- *Implementation costs* cover the direct costs of technology, software, and labor needed to integrate a new standard into your existing digital twin ecosystem.
- *Compliance and audit costs* are often the most overlooked cost. Many formal standards, especially those from ISO, require regular audits to maintain compliance. These audits can be costly, both in terms of fees and the internal labor hours spent on preparation, documentation, and the audit process itself.
- *Training and expertise* ensures your team has the necessary skills to work with a new standard and requires investment in training and can also mean hiring new, specialized personnel.